Stakeholders Primary Stakeholders • UC Students (End Users)	Use the system to coordinate group projects, tasks, and deadlines.
Secondary Stakeholders (Project Sponsor) • Aurek Chattopadhyay	Oversees project direction and evaluates outcomes.
Project Manager • Sri Chanakya Atreyapurapu	Plans sprints, manages scope, and ensures timely delivery.
Development Team Members • Tahreem Khan, Kingston Shorter, Sainidhi Sadaram, Arleen Monteiro	Design, implement, test, and deploy the system.
Instructor (Course Stakeholder) • Alex Pham	Evaluates compliance with software engineering principles and deliverables.

Functional Requirements (FR)

	User Management
FR1	The system shall allow students to create an account using a valid email.
FR2	The system shall allow registered users to log in and log out securely.
	Group Management
FR3	The system shall allow users to create study groups.
FR4	The system shall allow group creators to invite other users via invitation links or email.
FR5	The system shall allow users to accept or decline group invitations.
	Task & Deadline Management
FR6	The system shall allow users to create tasks within a group.
FR7	The system shall allow tasks to be assigned to one or more group members.
FR8	The system shall allow tasks to have due dates and priority levels
FR9	The system shall allow tasks to be moved between To-Do, In Progress, and Done states.
	Notifications
FR10	The system shall send reminders for upcoming deadlines.
FR11	The system shall notify users when tasks are assigned or updated.
	Collaboration & Tracking
FR12	The system shall display a shared task board visible to all group members.
FR13	The system shall track task completion status in real time.

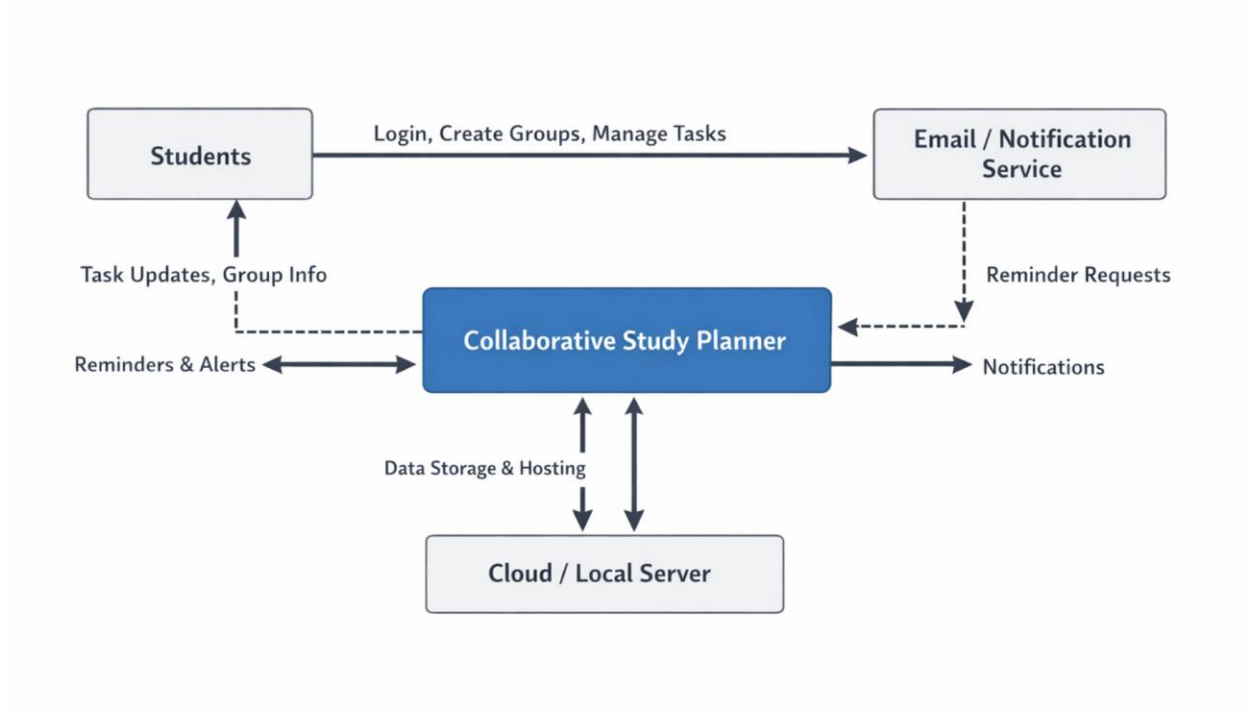
Non-Functional Requirements (NFR)

	Performance
NFR1	The system shall update task changes for all group members within 2 seconds.
	Usability
NFR4	The system shall be usable by first-time users without formal training.
NFR3	The user interface shall be accessible via modern web browsers.
	Reliability & Availability
NFR4	The system shall be available at least 95% of the time during the academic term.
	Security
NFR5	User authentication credentials shall be securely stored.
NFR6	Only authorized group members shall access group data.
	Scalability
NFR7	The system shall support at least 100 concurrent users without performance degradation.
	Maintainability
NFR8	The system shall follow modular design to support future feature expansion

Requirements Prioritization

High Priority	• User login and authentication • Group creation and invitations • Task creation, assignment, and deadlines • Shared task board
Medium Priority	• Automated reminders • Task status tracking • UI improvements
Low Priority	• CI/CD visualization • Deployment optimization • Advanced analytics

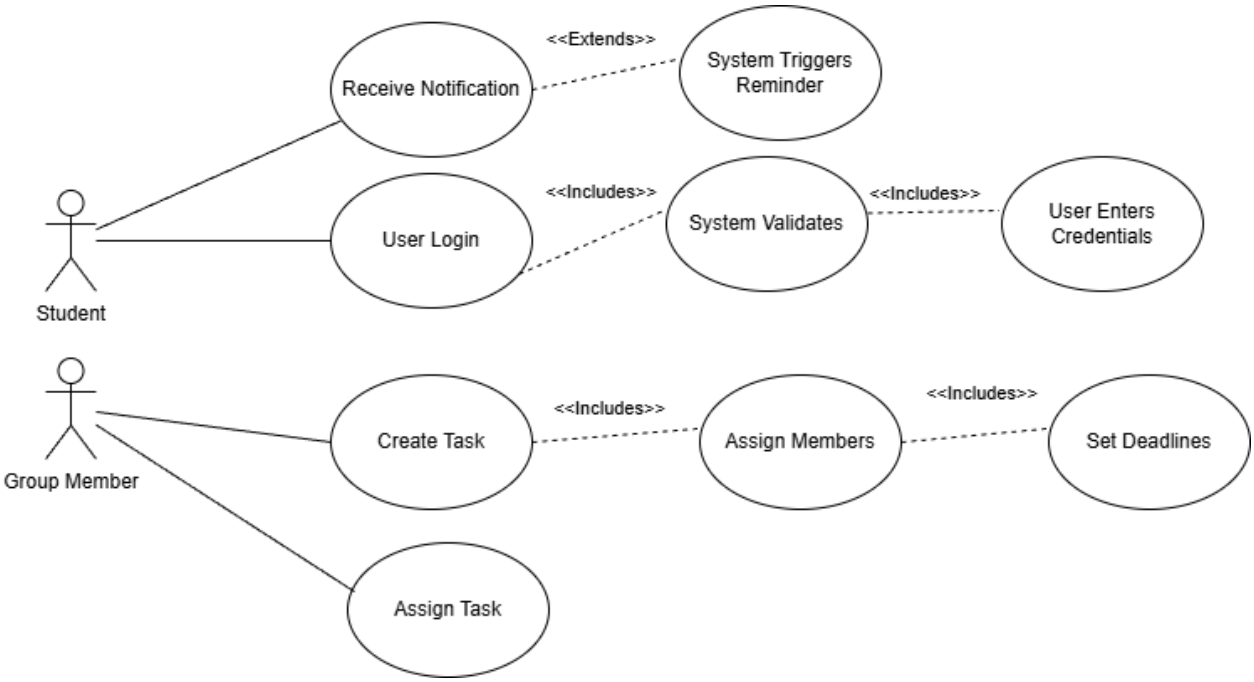
Context Diagram



Scenarios (Minimum 3)

Scenario 1: Creating a Study Group	• **Starting Situation:** Group members are enrolled in a shared project. • **Normal Flow:** Tasks are added, assigned, and moved across task states. • **What Can Go Wrong:** A task is assigned to the wrong user. • **Concurrent Activities:** Notifications are being sent. • **Ending State:** Task statuses accurately reflect progress.
Scenario 2: Managing Group Tasks	• **Starting Situation:** Group members are enrolled in a shared project. • **Normal Flow:** Tasks are added, assigned, and moved across task states. • **What Can Go Wrong:** A task is assigned to the wrong user. • **Concurrent Activities:** Notifications are being sent. • **Ending State:** Task statuses accurately reflect progress.
Scenario 3: Receiving Deadline Reminders	• **Starting Situation:** A task deadline is approaching. • **Normal Flow:** The system sends a reminderm notification. • **What Can Go Wrong:** Notification service is temporarily unavailable. • **Concurrent Activities:** Other tasks are being updated. • **Ending State:** User is reminded or retry is scheduled.

Use Cases (Minimum 3)

Use Case 1: User Login	• **Actor:** Student • **Precondition:** User has an account. • **Main Flow:** User enters credentials → system validates → user gains access. • **Postcondition:** User is logged in.
Use Case 2: Create and Assign Task	• **Actor:** Group Member • **Precondition:** User is part of a group. • **Main Flow:** User creates task → assigns member → sets deadline. • **Postcondition:** Task appears on shared board.
Use Case 3: Receive Notification	• **Actor:** Student • **Precondition:** Task deadline is near. • **Main Flow:** System triggers reminder → user receives alerts. • **Postcondition:** User is informed of upcoming deadlines.

Use Case Diagram